THE COLLEGE OF WOOSTER

The Department of Chemistry

CHEM 401: Introduction to Independent Study

Computational Identification of Unknown Organic Contaminants and Molecules in Remote Environments via Molecular Networking

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1 Summary

The widespread distribution of anthropogenic organic contaminants in remote high-altitude environments highlights a critical gap in our understanding of global chemical pollution. Compounds originating from industrial activity, agriculture, and consumer products can undergo long-range atmospheric transport and accumulate in regions such as the Himalayas, the Qinghai–Tibetan Plateau, and Mount Kilimanjaro. While previous studies have identified select contaminants in these environments, a large fraction of detected chemical features remain uncharacterized due to limitations in traditional targeted analytical approaches. This project addresses this gap by asking: to what extent can non-targeted, high-resolution mass spectrometry approaches expand the identification of unknown organic contaminants in remote surface waters?

The central hypothesis of this study is that a significant portion of unidentified molecular features detected in environmental samples can be identified or tentatively identified using non-targeted analysis combined with molecular networking. The goal is to identify both known contaminants and previously unannotated molecular features, thereby extending current environmental chemical databases rather than fully characterizing every compound.

To achieve this, surface water samples from remote high-altitude regions will be analyzed using liquid chromatography coupled with high-resolution mass spectrometry (LC-HRMS). Data acquisition will employ iterative data-dependent acquisition (DDA) to maximize MS/MS spectral coverage across both high- and low-abundance compounds. Resulting data will be processed using feature detection and alignment workflows to generate a comprehensive set of molecular features.

Identification will proceed in two stages. First, suspect screening will be performed by matching detected features against curated chemical databases using accurate mass, MS/MS fragmentation patterns, and retention time information to identify known contaminants. Second, non-targeted analysis will be applied to features that remain unidentified. Molecular networking based on MS/MS spectral similarity (cosine similarity scoring) will be used to group structurally related compounds, enabling unknown features to be linked to known chemical classes. Structural interpretation will leverage mass differences and fragmentation behavior to support tentative identification of these unknown compounds.

This integrated approach provides a more comprehensive framework for identifying contami-

nants in remote environments by combining database-driven methods with computational analysis of unknowns. The results will improve detection coverage, expand existing contaminant databases, and strengthen our understanding of global contaminant transport into sensitive high-altitude ecosystems.

2 Background and Significance

2.1 Environmental Contaminants

The pervasive and increasing presence of anthropogenic chemicals in the environment represents a major global challenge, requiring advanced analytical approaches to understand their distribution and impact. Human activities, including industrial processes, agriculture, and consumer product use, continuously release a wide range of chemicals such as pesticides, pharmaceuticals, industrial compounds, and plasticizers into the environment.[1–3] Many of these compounds are persistent and mobile, allowing them to spread far beyond their original sources and become globally distributed contaminants.

As a result, these chemicals have been detected even in remote ecosystems, including high-altitude regions such as Mount Everest, Kilimanjaro, and the Qinghai–Tibetan Plateau.[4–7] The presence of anthropogenic markers such as caffeine, pharmaceuticals, and industrial additives in these environments highlights the extent of human impact on global chemical cycles.[8,9]

The global spread of these contaminants poses significant ecological risks. High-altitude environments are particularly vulnerable due to their unique climatic and environmental conditions. Snow plays a critical role in these systems, acting as an efficient scavenger of atmospheric pollutants by capturing both gas-phase and particle-bound compounds.[10] Additionally, low temperatures in these regions slow chemical degradation, allowing contaminants to persist and accumulate over time.[4] Upon snowmelt, these accumulated contaminants can be released into surface waters, potentially impacting fragile alpine ecosystems and downstream environments.[11]

2.2 The Global Problem: Widespread Dispersion of Anthropogenic Chemicals

The continuous release of anthropogenic chemicals has resulted in their widespread presence across the globe. Many of these compounds, including persistent organic pollutants (POPs) and emerging contaminants, can undergo long-range atmospheric transport (LRAT), allowing them to reach remote regions far from their sources.[4,5] This process contributes to the global distribution of diverse chemical classes, including pesticides, pharmaceuticals, polycyclic aromatic hydrocarbons (PAHs), and plasticizers.

Previous environmental studies using non-targeted analysis (NTA) have demonstrated the presence of thousands of chemical features in environmental samples, highlighting the complexity of environmental chemical mixtures.[1,12] For example, HRMS-based studies have identified pesticides and pharmaceuticals in precipitation and snow samples, demonstrating that atmospheric transport is a major pathway for contaminant deposition.[8,9]

The widespread occurrence of these compounds contributes to the environmental "exposome," which represents the totality of environmental exposures experienced over time.[1] However, a large fraction of detected chemical features remains unidentified due to limitations in existing chemical databases, emphasizing the need for improved analytical and computational approaches.

2.3 Why Remote Peaks? Kilimanjaro and the Himalayas as "Cold Traps

Remote high-altitude environments such as the Himalayas and the Qinghai–Tibetan Plateau serve as critical locations for studying global contaminant transport. Because these regions lack significant local pollution sources, the detection of contaminants provides strong evidence of long-range atmospheric transport .[4,6]

These environments act as "cold traps," where semi-volatile organic compounds (SVOCs) condense and accumulate due to decreasing temperatures with altitude.[4,10] This process, often referred to as global or altitudinal fractionation, results in the enrichment of certain contaminants at higher elevations. The combination of low temperatures, high precipitation (especially snow), and complex terrain enhances deposition and retention of contaminants in these regions[7,10] as seen in Figure 1.

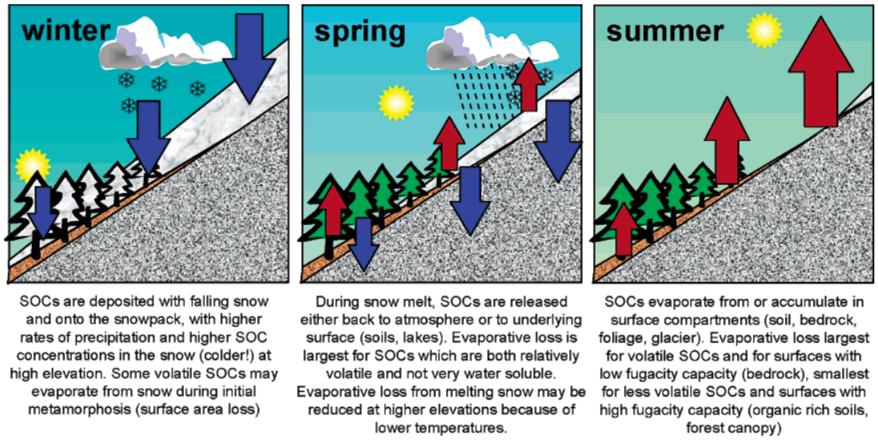


Figure 1: Air-surface exchange of semi-volatile organic contaminants (SVOCs) along a temperate mountain slope across different seasons.[4]

Previous studies have demonstrated the presence of persistent contaminants such as organochlorine pesticides and industrial chemicals in high-altitude environments as seen in Figure 2.[4,5] These findings highlight the importance of high-altitude ecosystems as reservoirs of global pollution and emphasize the need for continued monitoring.

Despite this, there remain significant gaps in our understanding of organic contaminants in regions such as the Qinghai–Tibetan Plateau, particularly with respect to unknown and emerging compounds.[6] This gap motivates the need for advanced analytical approaches capable of detecting and identifying a broader range of contaminants.

chemical	location (elevation[a])	date	ref
	OPPs and Other Currently Used Pesticides		
chlorpyrifos, diazinon, parathion	3 elevations (114/200, 533, 1920 m) in and below Sequoia National Park, Sierra Nevada, CA	15 days (Nov. 1990−Feb. 1991)	*26*
chlorpyrifos, methidathion		8 days (May−Oct. 1994)	*27*
chlorpyrifos, diazinon, malathion, chlorothalonil, trifluralin, endosulfan		5 days (May−Sept. 1996)	*28*
	SOCs		
PCBs in 2 HiVol samples	Lake Tahoe, Sierra Nevada, CA, USA (ca. 1900 m)	July 1995	*29*
OCPs in 7 HiVol samples	Bow Lake, Rocky Mtns., Canada (1940 m)	summer 1998	*61*
PAHs in 36 HiVol samples	Estany Redó, Pyrenees (2240 m)	1996−1997	*34*
	Gossenköllesee, Austrian Alps (2413 m)	1996−1997	
	Øvre Neådalsvatn, Norway (728 m)	1998	
PCBs and OCPs in 23 HiVol samples	Bow Lake (1974 m), Donald Station (770 m), Lower Kananaskis Lake (1667 m), Sundre, Rocky Mountains, Canada (1115 m)	June 1999−Aug. 2000	*33*
PCBs and OCPs in 20 HiVol samples	Pico de Teide, Canary Islands (47 and 2367 m)	June 1999−July 2000	*25*
α-HCH, HCB, DDT	Yahiko Mtn. (634 m), Japan	May−Nov. 2001	*30*
OCPs in 8 passive samples	Rocky Mountains, Canada (800−2300 m)	2000−2001 (annual average)	*31, 32*
PCBs and OCPs in 30 HiVol samples	Estany Redon, Central Pyrenees (2240 m)	Nov. 2000−Jan. 2003	*106*
	Skalnate Pleso, High Tatras (1778 m)		
OCPs in 6 HiVol samples	Dingri (4340 m), Rongbu Valley (5050 m), Tibetan Plateau	May−June 2002	*16*

[a] Elevation given in meters above sea level.

Figure 2: Organic contaminant concentrations in mountain air measured by Daly et al. (2005).[4]

2.4 Atmospheric Chemistry and Transport Mechanisms

Organic contaminants enter the atmosphere through processes such as volatilization, combustion, and agricultural application. Once airborne, these compounds can undergo chemical transformations, including oxidation and photolysis, which influence their persistence and environmental behavior.[13]

The long-range transport of these contaminants is often described by the "grasshopper effect," in which compounds repeatedly volatilize, travel through the atmosphere, and redeposit in cooler regions. This process enables contaminants to travel thousands of kilometers from their original sources and accumulate in high-altitude and polar environments.[4,5,7]

Mountain regions further enhance contaminant deposition through several mechanisms, including:

- Orographic precipitation, which increases scavenging of airborne pollutants.[7]
- Snow deposition, which efficiently captures contaminants.[10]
- Temperature gradients, which drive cold trapping and chemical partitioning as seen in Figure 3.

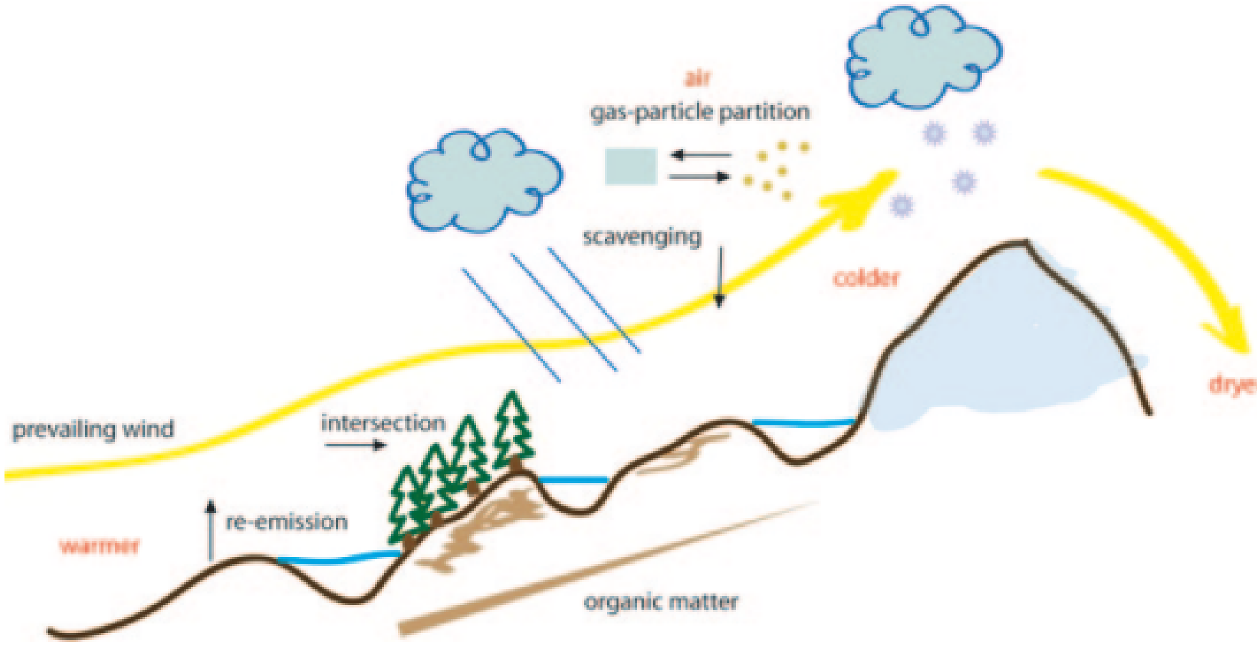


Figure 3: Transport mechanisms of atmospheric contaminants in mountainous regions and what physical factors drive the accumulation of these pollutants.[7]

These processes make high-altitude environments particularly sensitive indicators of global contaminant transport and transformation.

2.5 Significance: Advancing Identification Beyond Traditional Targeted Approaches

Traditional environmental analysis has relied heavily on targeted methods, which focus on a predefined list of known contaminants. While effective for monitoring regulated compounds, these approaches fail to capture the full diversity of chemicals present in environmental samples.[1]

High-resolution mass spectrometry (HRMS), combined with non-targeted analysis (NTA) and suspect screening, has emerged as a powerful approach for detecting a wide range of known and unknown compounds in complex environmental matrices.[12,14] These methods enable the detection of thousands of molecular features in a single sample, many of which cannot be matched to existing databases.[14]

Previous studies have demonstrated the effectiveness of HRMS-based NTA in identifying contaminants such as pharmaceuticals, pesticides, and industrial chemicals in environmental samples.[8,9] However, a significant limitation remains: many detected features cannot be confidently identified due to incomplete reference libraries.

To address this challenge, this project applies molecular networking as a complementary computational approach to extend identification beyond existing databases. Molecular networking groups compounds based on MS/MS similarity, enabling structurally related molecules to be linked even in the absence of direct database matches.[15–17] This approach allows for the propagation of annotations and the identification of "known unknowns," significantly expanding the scope of contaminant identification.

The significance of this research lies in its ability to improve contaminant identification in remote environments, generate more comprehensive chemical profiles, and address critical gaps in current environmental monitoring frameworks. By combining advanced HRMS workflows with molecular networking, this study contributes to a more complete understanding of global contaminant distribution and supports future efforts in environmental risk assessment and policy development.

3 Specific Aims

3.1 Goal

This project aims to identify organic contaminants in surface water samples from remote high-altitude environments using high-resolution mass spectrometry (HRMS). Remote environments such as high-altitude regions are known to accumulate contaminants through long-range atmospheric transport, making them important systems for studying global pollutant distribution.[4,5,7] While database-driven approaches enable identification of known contaminants, many environmental samples contain complex mixtures that are not fully represented in existing libraries, resulting in substantial gaps in identification coverage.[1,12] I hypothesize that a significant fraction of these uncharacterized molecules can be annotated and assigned putative identities through non-targeted analysis, particularly molecular networking, which enables structurally related compounds to be grouped based on MS/MS similarity even in the absence of direct database matches.[15,16] Together, these approaches will improve contaminant identification and extend existing chemical databases for remote environmental systems.

3.2 Aim 1: Comprehensive MS/MS Data Acquisition

The first aim is to generate a comprehensive and high-quality MS/MS dataset from surface water samples using liquid chromatography coupled with high-resolution mass spectrometry (LC-HRMS). High-resolution mass spectrometry is widely used in environmental non-targeted analysis due to its ability to detect a broad range of known and unknown compounds with high mass accuracy.[14,18] Chromatographic separation will be performed using a C18 reverse-phase column with gradient elution to separate compounds across a range of polarities, improving detection of chemically diverse contaminants. Samples will be analyzed in both positive and negative electrospray ionization (ESI) modes to maximize compound coverage across different ionization behaviors.[19]

Data acquisition will be performed using an iterative data-dependent acquisition (DDA) strategy, in which each sample is analyzed across multiple runs with dynamically updated exclusion lists to increase fragmentation coverage of detected features.[20] This approach improves detection of low-abundance compounds and enhances overall MS/MS coverage in complex environmental mixtures. High-quality data acquisition is a critical foundation for non-targeted identification workflows and directly impacts downstream annotation performance.[12] The outcome of this aim will be a comprehensive feature-rich dataset suitable for contaminant identification.

3.3 Aim 2: Identification of Known Contaminants Using Existing Library

The second aim is to identify known contaminants of emerging concern (CECs) using suspect screening approaches based on existing chemical databases (EPA contaminant database). Non-targeted and suspect screening workflows are widely used to assess environmental contaminants in complex matrices such as water and snow samples.[8,9] Processed HRMS data will be analyzed using software such as MS-DIAL for feature detection, alignment, and spectral matching.[21] Compound identification will rely on accurate mass measurements, MS/MS spectral similarity, isotopic patterns, and retention time as complementary parameters for annotation.[22]

Confidence in compound identification will be evaluated using the Schymanski framework, which provides standardized levels of identification certainty in non-targeted analysis.[23] This approach enables systematic identification of known contaminants while highlighting gaps in existing databases and prioritizing features for further analysis.[1,24] The outcome of this aim will establish a baseline profile of confidently annotated known contaminants.

3.4 Aim 3: Extension of Contaminant Identification Beyond Current Databases Through Molecular Networking

The third aim is to extend contaminant identification beyond existing databases by applying molecular networking to assign putative identities to unknown features. Molecular networking is a computational approach that groups MS/MS spectra based on fragmentation similarity and has been widely applied for large-scale analysis of complex chemical datasets.[15] Spectral similarity will be quantified using cosine similarity scoring, a standard method for comparing MS/MS spectra and constructing molecular networks.[16,17] Preprocessing steps such as spectral filtering and alignment improve the reliability of similarity scoring and network construction.[25,26]

Rather than full structural characterization, this approach will focus on propagating annotations and assigning likely structural relationships between unknown and known compounds within molecular networks. Transformation products are a major source of unknown chemical features in environmental systems and are often not captured in existing databases, contributing to gaps in identification.[6,13] By leveraging spectral similarity and mass difference relationships, this aim will enable identification of previously unannotated compounds and provide insight into contaminant transformation patterns.

The outcome of this aim will be an expanded set of annotated and putatively identified contaminants, addressing key limitations in current non-targeted analysis workflows and improving coverage of environmental chemical space.

4 Research Design and Methods

4.1 Hypothesis

This project aims to identify contaminants of emerging concern (CECs) in surface water samples from remote, high-altitude environments, including Mt. Everest, the Tibetan Plateau, Qinghai Lake, and Mt. Kilimanjaro, using high-resolution mass spectrometry (HRMS). While existing database-driven, targeted approaches can successfully identify known contaminants, most current software and reference libraries are not well optimized for complex environmental matrices, leaving significant gaps in identification coverage. I hypothesize that a substantial fraction of these unidentified molecules can be assigned through non-targeted analysis, particularly molecular networking, thereby extending existing contaminant databases. Together, these approaches will improve contaminant identification in remote environmental systems where current reference frameworks are incomplete.

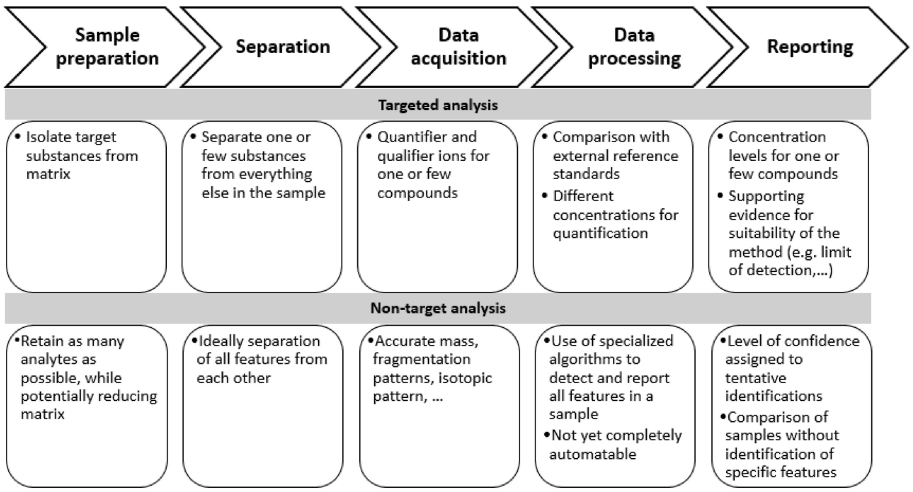


Figure 4: Comparison of non-targeted analysis (NTA) and targeted analysis (TA) approaches for identification of organic contaminants.[12]

4.2 High-Resolution Mass Spectrometry Data Acquisition

All samples will be analyzed using liquid chromatography coupled with high-resolution mass spectrometry (LC-HRMS) using an Agilent 1260 Infinity II HPLC and Agilent 6546 QTOF-MS. Chromatographic separation will be performed using a C18 reverse-phase column, allowing for the retention and separation of semi-polar to nonpolar organic compounds.

LCMS-grade water and LCMS-grade methanol will be used for all mobile phases to ensure minimal background contamination. Data will be acquired in both positive and negative electrospray ionization (ESI) modes to maximize compound coverage. For positive ionization mode, the mobile phase will consist of 0.1% formic acid in LCMS-grade water (aqueous phase) and methanol with 0.1% formic acid (organic phase).[19] For negative ionization mode, the mobile phase will consist of 10 mM ammonium acetate in LCMS-grade water (aqueous phase) and 10 mM ammonium acetate in methanol (organic phase).[19] Gradient elution will be used in both modes to separate compounds across a range of polarities.

The analytical workflow will begin with MS1 full-scan acquisition to detect all ionizable features present in the samples. From these data, exclusion and inclusion lists will be generated (Figure 5). The exclusion list will contain background contaminants high-abundance ions that do not require repeated fragmentation, while the inclusion list will prioritize features of interest, particularly those matching entries in existing chemical databases.[20]

Subsequent analyses will employ MS/MS acquisition using an iterative data-dependent acquisition (DDA) strategy. In DDA, precursor ions are selected in real time after a MS-only scan based on their sufficiently high intensity for fragmentation. In this study, each sample will be analyzed across 3–4 iterative runs. After each run, previously fragmented ions are added to an exclusion list, allowing subsequent runs to target new, lower-abundance features. This iterative process increases MS/MS coverage by systematically expanding fragmentation across the detected feature space, improving the likelihood of identifying both known and unknown compounds in complex environmental mixtures.

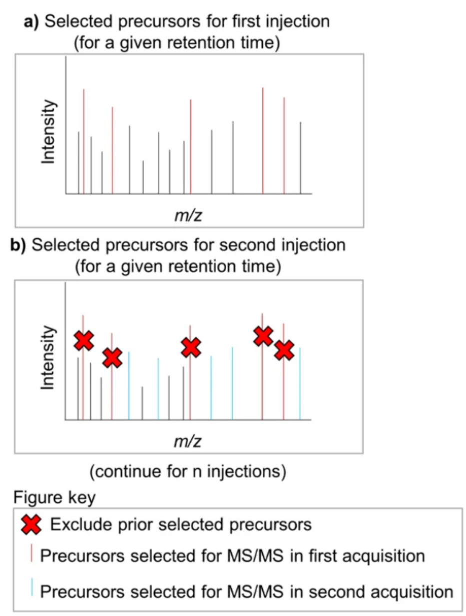


Figure 5: Example of exclusion and inclusion of features to establish our lists where you exclude prior selected precursor ions after each run.[20]

4.3 Data Processing and Feature Extraction

Raw LC-HRMS data will be processed using MS-DIAL v5.5.2505300. Data processing will include peak detection, alignment across samples, and feature annotation. A representative workflow is shown in Figure 6; although this figure illustrates data-independent acquisition (DIA), the overall processing steps are similar to those used for data-dependent acquisition (DDA) in this study.

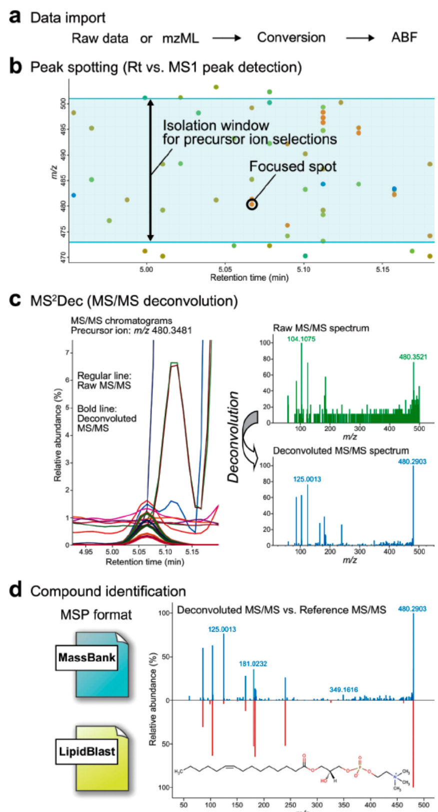


Figure 6: This workflow illustrates data independent acquisition (DIA) using MS Dial while different from our data dependent acquisition (DDA) we are seeking something similar.[21]

Detected signals will initially be treated as features till confirmed with a match by MS-DIAL which identifies and quantifies the features before assigning annotations to them or molecular networking using cosine similarity scoring, these features are defined by their mass-to-charge ratio (m/z), retention time, and associated MS/MS spectra. Moreover, a feature is only considered a confirmed compound once sufficient evidence supports its identification.

Quality control steps, including blank subtraction and filtering of low-confidence signals, will be applied to remove background contaminants.[12] The resulting feature table will represent the set of detected molecular features across all samples and will serve as the basis for downstream identification and analysis.

4.4 Suspect Screening for Known Contaminants

To identify contaminants of emerging concern (CECs), suspect screening will be performed using mass spectra databases, such as the Agilent environmental contaminant database. Figure 7 illustrates how MS/MS spectral databases are constructed and used in conjunction with LC-HRMS data to identify compounds based on fragmentation patterns.

Features will be matched to database entries using a combination of accurate mass, MS/MS spectral similarity, isotopic pattern, and retention time.[22] Retention time serves as an important orthogonal parameter that strengthens confidence in candidate matches.

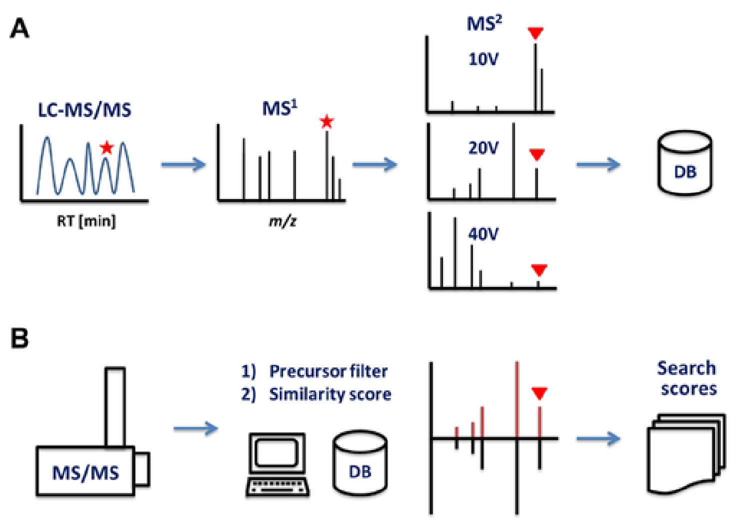


Figure 7: This figure illustrates the MS/MS database creation and how LCMS is used to identify compounds based on their fragmentation patterns.[22]

Candidate annotations will be evaluated using the Schymanski framework, which classifies identification confidence into levels based on available evidence (Figure 8).[23] The Schymanski levels reflect the degree of confidence in a proposed identification rather than representing definitive assignments. The usage of the levels approach enables identification of known contaminants while simultaneously highlighting gaps in existing chemical databases where features cannot be confidently matched.[1]

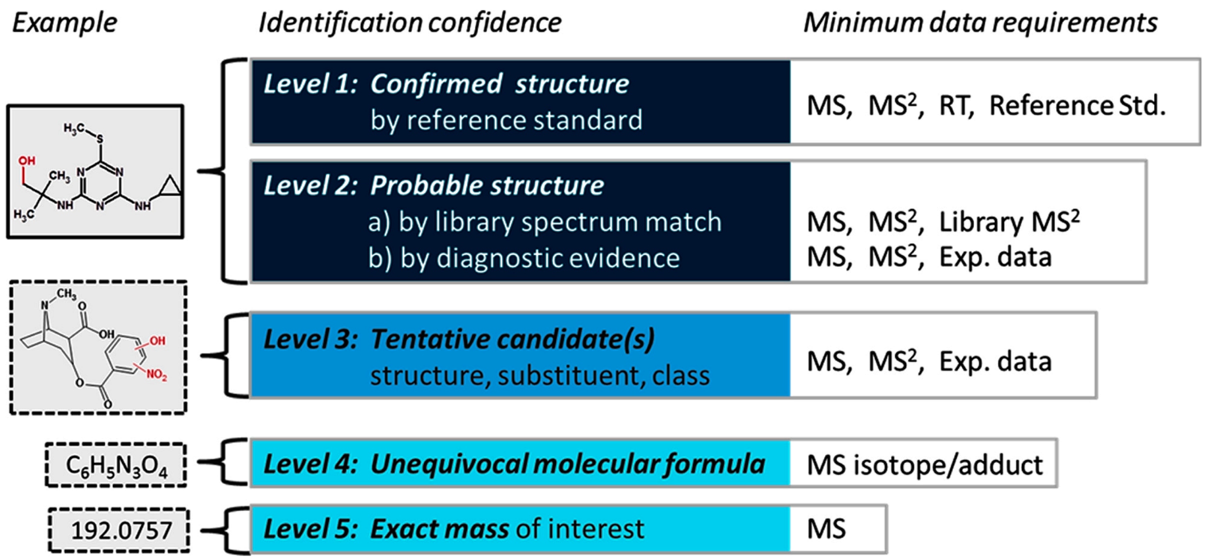


Figure 8: Confidence levels example for molecular feature identification based on the Schymanski framework.[23]

4.5 Molecular Networking and Non-Targeted Analysis

To extend beyond database-driven identification, molecular networking will be used to identify unknown features. Molecular networking is a computational approach that groups compounds based on similarities in their MS/MS fragmentation patterns, enabling structurally related molecules to be identified even when exact matches are not present in databases.[15]

Existing tools such as Global Natural Products Social Molecular Networking (GNPS) are widely used for this purpose; however, GNPS is primarily designed for natural product datasets and is not optimized for environmental contaminant analysis.[15,16] As shown in Figure 10 GNPS is the exact workflow we are seeking but can lead to gaps in annotation when applied to environmental samples. Therefore, a custom computational workflow will be developed to better suit the characteristics of environmental HRMS data.

Spectral similarity between molecular features will be evaluated using cosine similarity scoring of MS/MS fragmentation spectra.[17,27] Cosine similarity is a mathematical approach used to measure how similar two vectors are by calculating the cosine of the angle between them. For two vectors, x and y, the dot product is defined as:

$$x \cdot y = x_1 y_1 + x_2 y_2$$

This relationship can also be expressed geometrically as:

$$x \cdot y = |x||y| \cos(\theta)$$

where θ is the angle between the two vectors and $|x|$ and $|y|$ represent the magnitudes of the vectors. By rearranging the equation, the cosine similarity between the vectors becomes:

$$\text{similarity} = \cos(\theta) = \frac{x \cdot y}{|x||y|}$$

For positive vector values, cosine similarity ranges from 0 to 1:

- A value of 1 indicates identical vectors
- A value of 0 indicates completely unrelated or orthogonal vectors

This concept can be expanded to n-dimensional vectors:

$$\text{similarity} = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2}\sqrt{\sum_{i=1}^{n} y_i^2}}$$

In MS/MS spectral analysis, each fragmentation spectrum is represented as a vector composed of fragment ion peaks defined by their mass-to-charge ratio (m/z) and corresponding intensity values. To improve spectral comparison, fragment peaks are weighted according to both their intensity and m/z value using the weighted intensity function:

$$W_i = (m/z_i)^2 \sqrt{I_i}$$

where:

- m/z_i is the mass-to-charge ratio of the i^{th} fragment ion
- I_i is the intensity of the i^{th} fragment ion

This weighting increases the contribution of larger and more intense fragment ions to the overall similarity score. Two spectral vectors are then constructed:

- E: experimental spectrum from mass spectra data just like in Figure 9.

- D: database or reference spectrum from a mass spectral database just like in Figure 9.

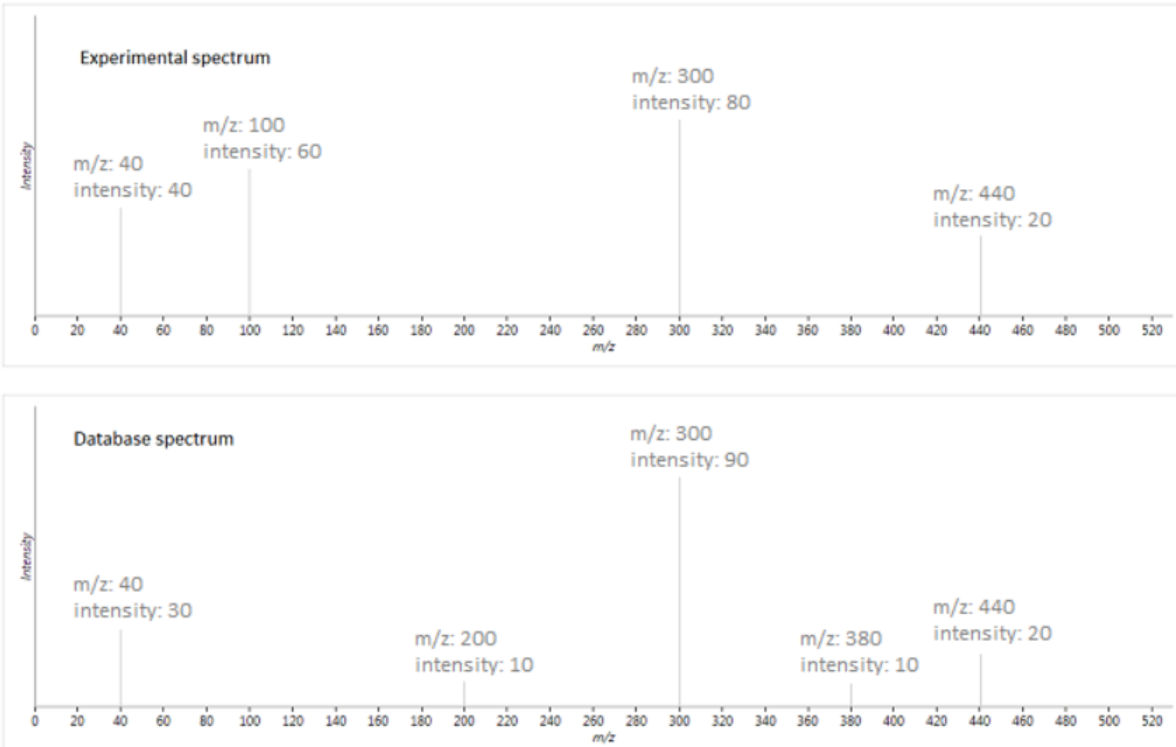


Figure 9: This example illustrates how the experimental spectrum (E) and database spectrum (D) are compared using cosine similarity scoring based on their fragment ion m/z values and intensities.[17,28]

All fragment ion m/z values from both spectra are combined and aligned in ascending order. For each aligned m/z value, one of three cases occurs:

1. An experimental peak exists without a matching database peak

2. A database peak exists without a matching experimental peak

3. Both spectra contain matching fragment ions within a specified mass tolerance

The vectors are populated as follows:

- Experimental-only peaks contribute a weighted value to E and 0 to D

- Database-only peaks contribute 0 to E and a weighted value to D

- Matching peaks contribute weighted values to both vectors

The cosine similarity score is then calculated using the normalized dot product of the two weighted vectors:

$$\text{Cosine Similarity} = \frac{\sum(A_i \cdot B_i)}{\sqrt{\sum(A_i^2)} \cdot \sqrt{\sum(B_i^2)}}$$

where A_i and B_i represent the weighted intensities of aligned fragment ions in the experimental and reference spectra, respectively.

This scoring method evaluates:

- The presence of shared fragment ions

- Relative fragment ion intensities

- Agreement of fragmentation patterns

- Penalties from unmatched peaks

Features exceeding 0.25 threshold will be grouped to construct molecular similarity networks, enabling the identification of structurally related unknown compounds and potential environmental transformation products. An example of this molecular networking workflow is illustrated in Figure 10. While the Global Natural Products Social Molecular Networking (GNPS) framework was originally designed for natural product datasets, a similar strategy will be adapted in this study for environmental HRMS analysis.

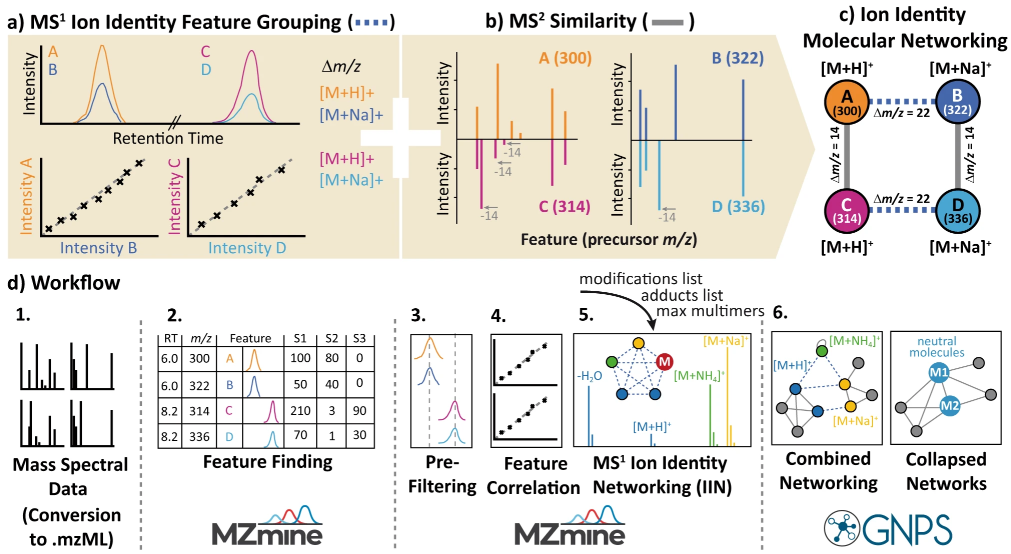


Figure 10: This example of Global Natural Product Social (GNPS) molecular networking illustrates the relationships between different molecular features and what is used to identify them based on their similarity to known compounds. It first connects and sorts ions, then annotates them based on their spectral similarities and finally proposes a structure.[16]

4.6 Structural Interpretation and Data Analysis

Structural interpretation will focus on analyzing mass differences and fragmentation patterns within molecular networks to propose tentative structures for unknown compounds.[13] Mass shifts are of particular interest because they often correspond to known environmental transformation processes, such as oxidation, reduction, or functional group modification. For example:

- A mass increase of +16 Da may indicate the addition of an oxygen atom, commonly associated with hydroxylation reactions.
- A mass shift of +14 Da may suggest methylation.
- A mass change of +16 Da combined with loss of 2 hydrogen atoms may indicate the formation of a carbonyl group through oxidation.

These transformation processes are well-documented in environmental chemistry and provide insight into degradation pathways and secondary pollutant formation.[6,13] Fragmentation patterns will be analyzed alongside these mass differences to strengthen structural hypotheses.[16] A feature-based molecular networking framework (Figure 11) will be used to visualize relationships between features and support the identification of unknown compounds based on their similarity to known structures.

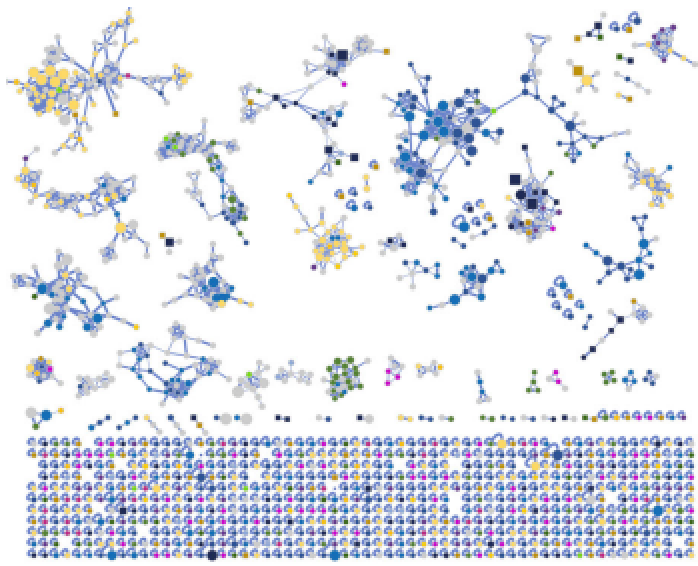


Figure 11: This feature based graph shows how ions become connected in a molecular network based on their similarity, facilitating the identification of unknown compounds.[16] The colors in the graph are by ion identities.

This integrated workflow as seen in Figure 12 enables (1) comprehensive MS/MS data acquisition, (2) identification of known contaminants using existing libraries, and (3) extension beyond current databases through molecular networking, directly supporting the three specific aims of this study.

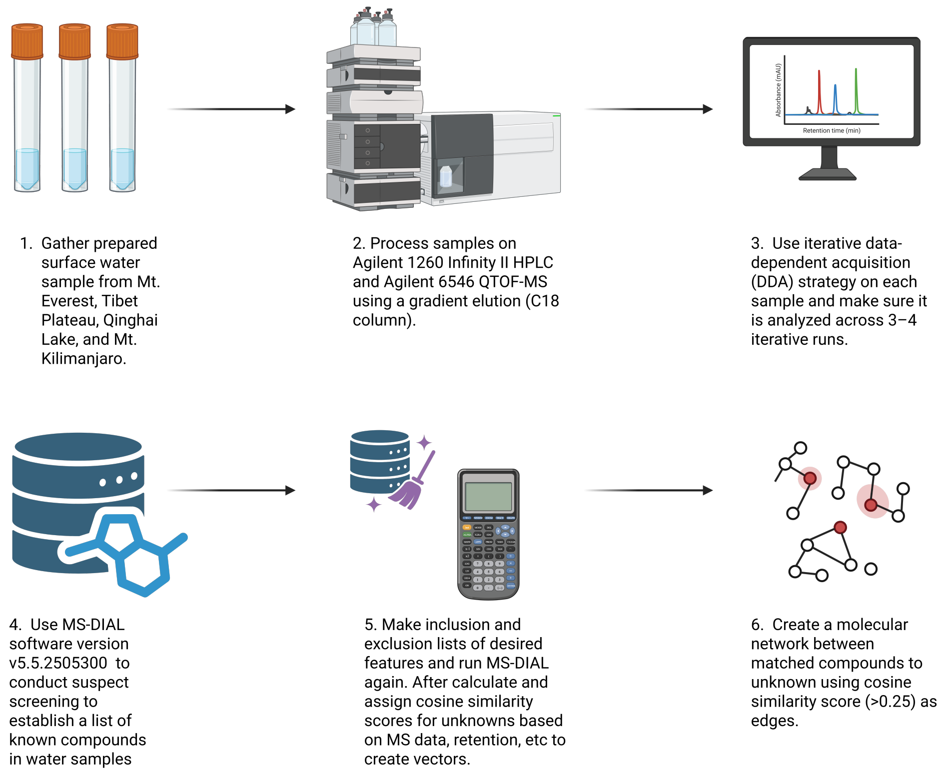
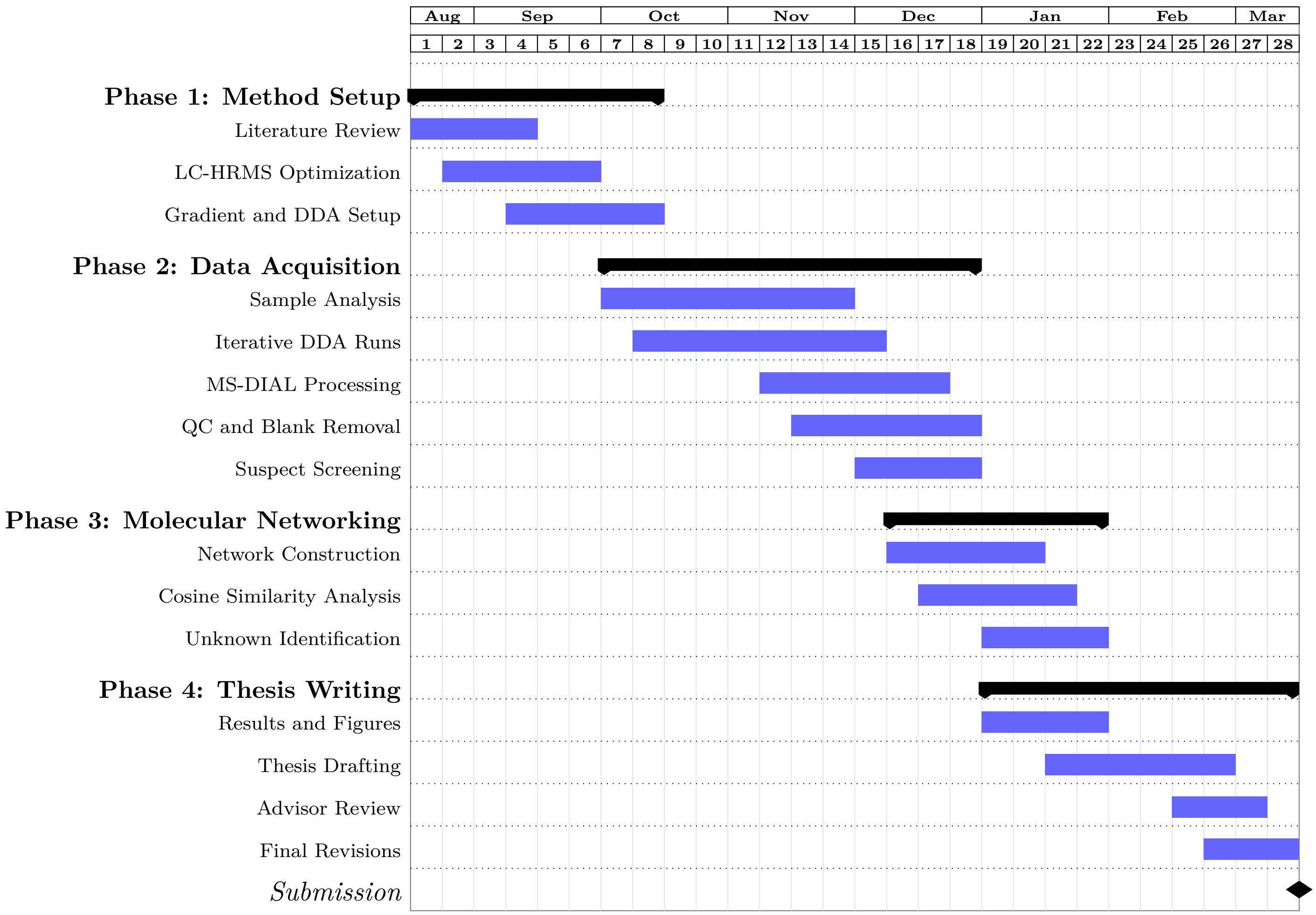


Figure 12: The full workflow of this project, illustrating how the different components of data acquisition, processing, and analysis are integrated to achieve the research objectives.

5 Projected Timeline and Budget

5.1 Timeline



5.2 Budget

Table 1: Project Budget: Chemicals and Instrumentation

Chemical/Instrument	Source	Location	Cost	Quantity
Agilent 1260 Infinity II HPLC	Department Instrumentation	Instrument Room	$0.00	1
Agilent 6546 QTOF-MS	Department Instrumentation	Instrument Room	$0.00	1
LC Column (C18 Reverse Phase)	Sigma-Aldrich	Instrument Room	$0.00	1
LCMS Grade Water	VWR	Chemical Storage	$120.00	4
LCMS Grade Methanol	Sigma-Aldrich	Chemical Storage	$150.00	4
Formic Acid (Optima LC/MS Grade)	Fisher Scientific	Chemical Storage	$65.00	2
Ammonium Acetate (LCMS Grade)	Sigma-Aldrich	Chemical Storage	$85.00	2
MS-DIAL Software	Open-Source Software	Laboratory Computers	$0.00	1
GNPS Molecular Networking Platform	GNPS	Laboratory Computers	$0.00	1
Total			**$775.00**	

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